

# **A Hand Movement Capture and Replay System for Subject Guidance in Motor Studies Using a Realistic 3D Hand Model and Camera Recordings**

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Informatics : Games Engineering  
Bachelor Thesis  
2026**



# Motivation and Problem Statement

Create Hand Instructions  
in 3D Environments



Requires Specific Understandings of  
Animations and Inverse Kinematics



Difficult for People without  
Prior Experience

# Motivation and Problem Statement

Simple Way to Create  
Instructions



[Video to show]

No Need for Specific Knowledge



Motion Capture

# Motivation and Problem Statement

MediaPipe Data is  
Inconsistent



Need Filtering for High  
Quality Results

# Motivation and Problem Statement





# Goals

1. Create a Complete Pipeline



Create Hand Instructions with only Camera Recordings



No Direct Interaction with Animation System

2. Find a Suitable Filtering Methods



[Mention IK]

Reduce Jitter and Noise



Output High Quality Hand Instructions



# Goals

- Convert recorded hand motion video into hand movements instructions
- Achieve realistic motions in 3D environments by reducing jitter, noise from MediaPipe
- [Create a complete pipeline where ....] No prior knowledge required in animation system, ease to use



# State of Science and Technology

**Motion Tracking Devices**

**Filtering Methods - Kalman filter and One Euro filter**

**Evaluation Metrics - MDCSS**

**Unreal Engine 5 and Metahuman Selection**





# State of Science and Technology - Motion Tracking Devices

## Motion Tracking Devices

[Images]

[RGB depth camera]



# State of Science and Technology.

Motion Tracking Devices

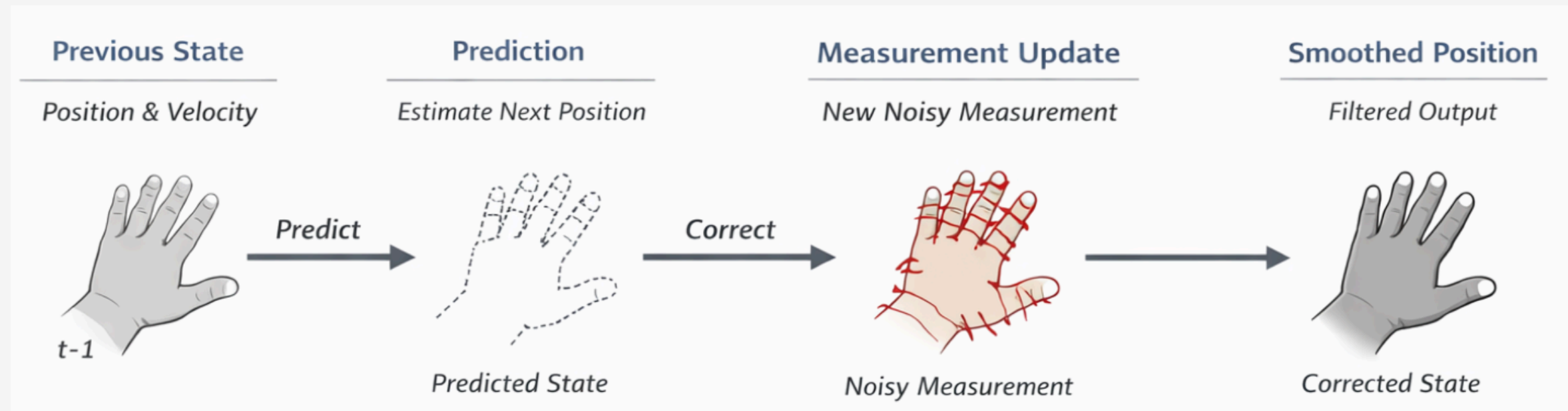
Filtering Methods - Kalman filter and One Euro filter

Evaluation Metrics - MDCSS

Unreal Engine 5

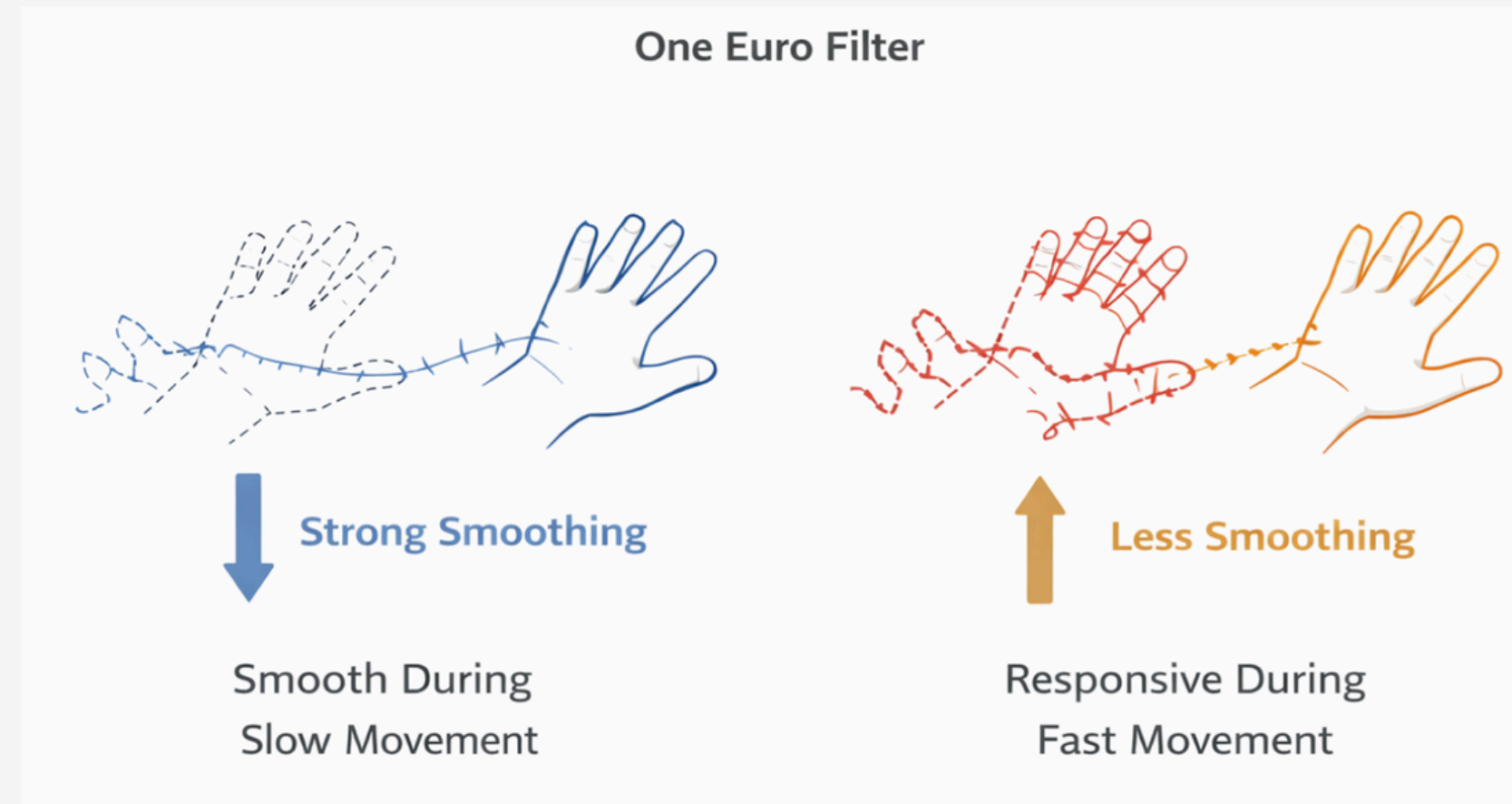


# Methodology - Standard Linear Kalman Filter



- Recursively estimate position
- Reduce noise by predicting next position

# Methodology - One Euro Filter



## Core Concept:

- Slow movements are smoothed strongly to reduce jitter
- Fast movements are smoothed less to preserve responsiveness

# State of Science and Technology.

Motion Tracking Devices

Filtering Methods - Kalman filter and One Euro filter

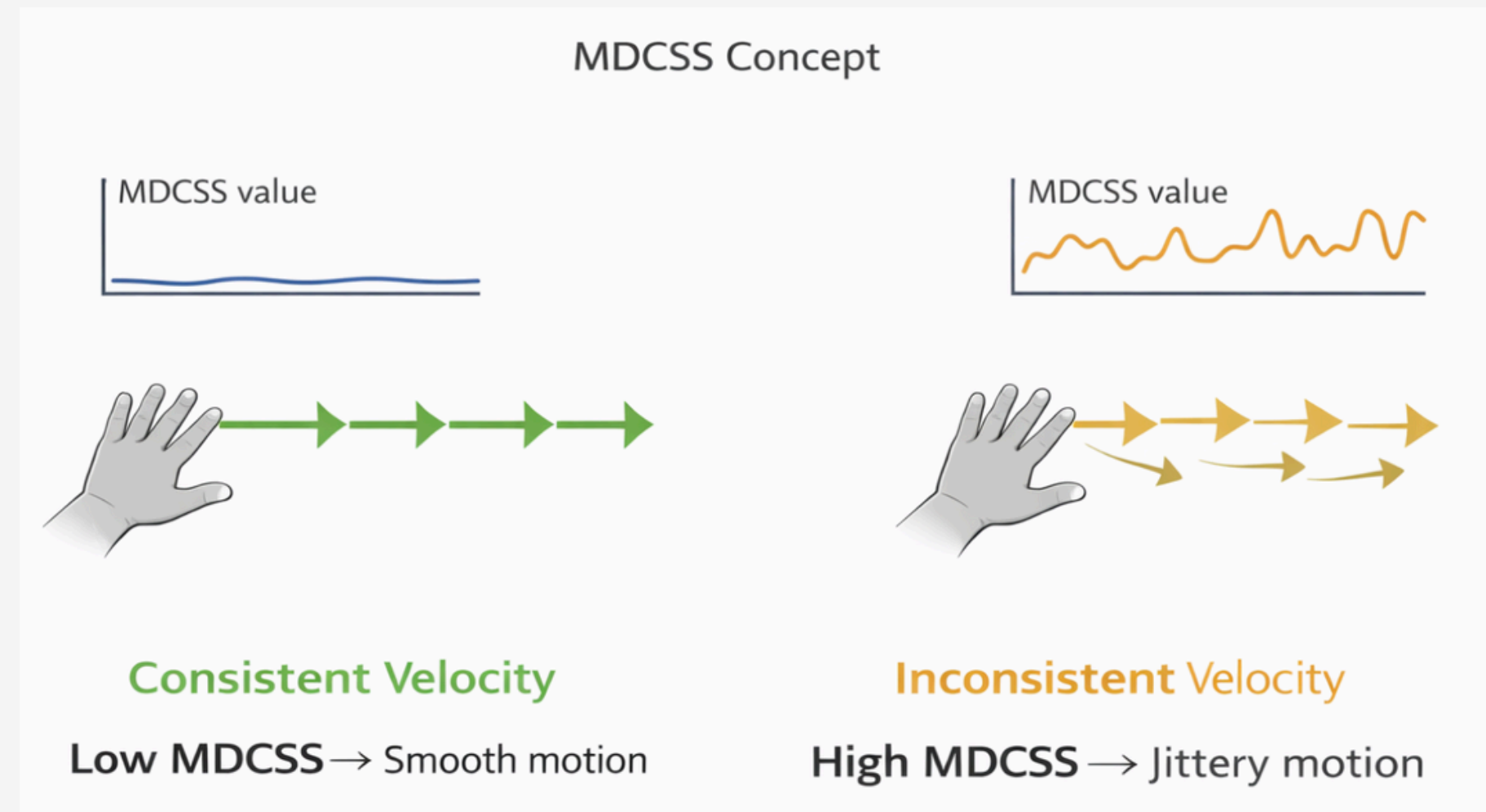
Evaluation Metrics - MDCSS

Unreal Engine 5





# Movement Dynamics Clutter Spectrum Strength



Detect jitter by calculating direction of movement each frame

# State of Science and Technology.

Motion Tracking Devices

Filtering Methods - Kalman filter and One Euro filter

Evaluation Metrics - MDCSS

Unreal Engine 5 and Metahuman Selection



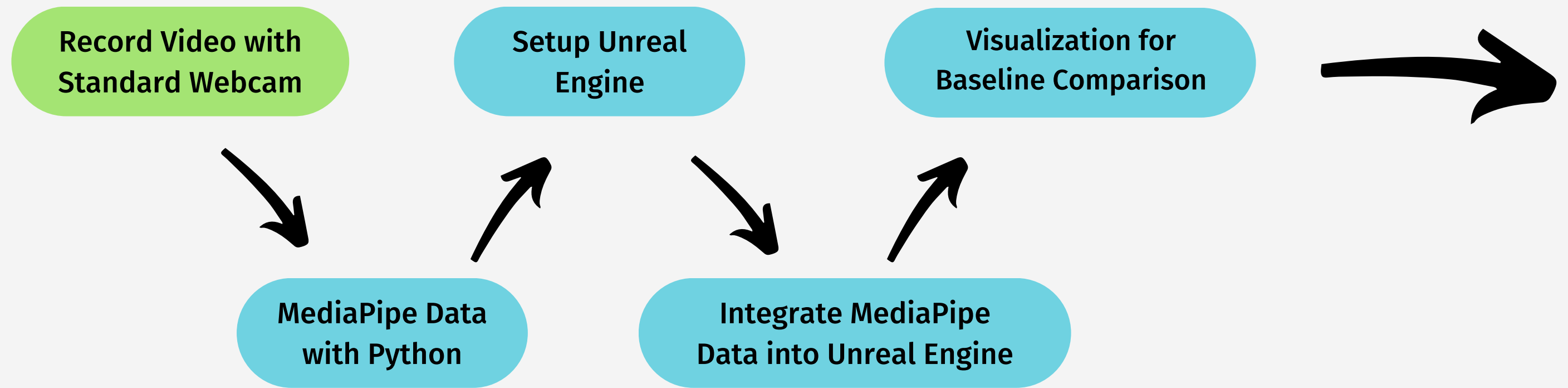
# State of Science and Technology.

## Game Engine: Unreal Engine 5 and Metahuman Selection

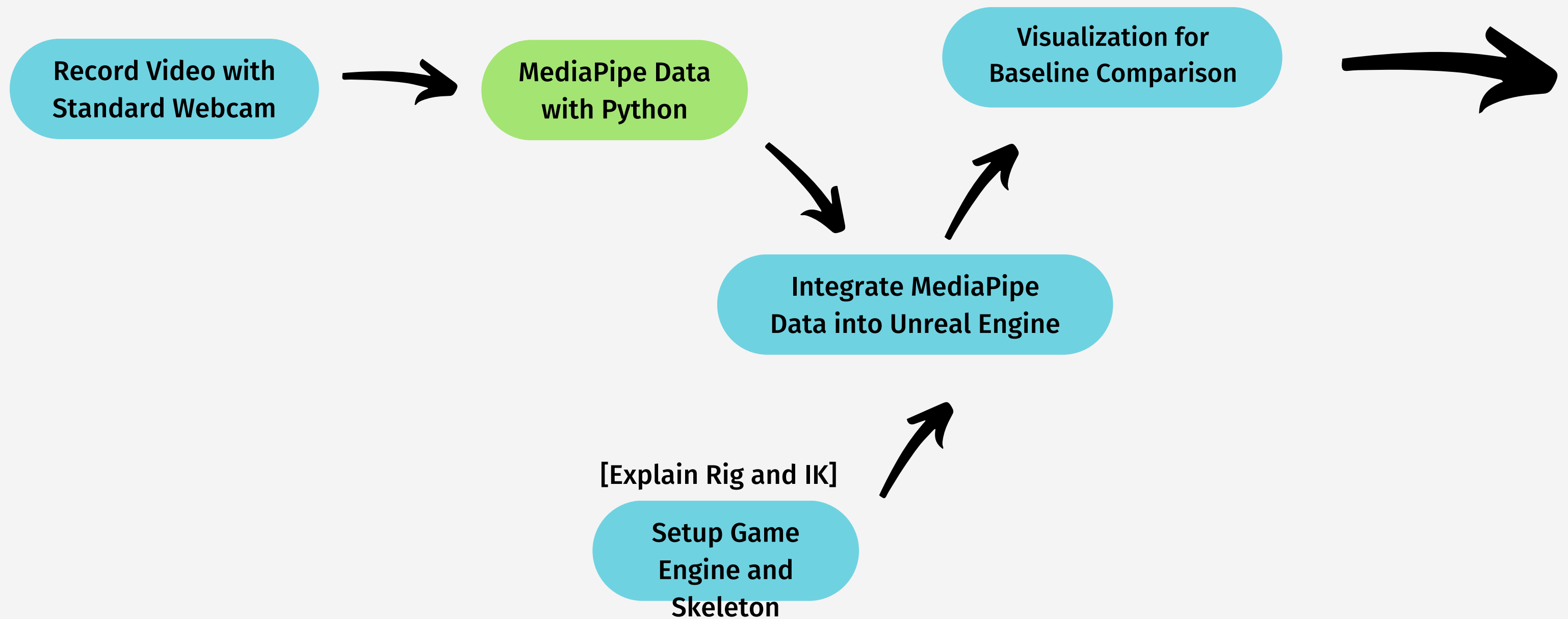




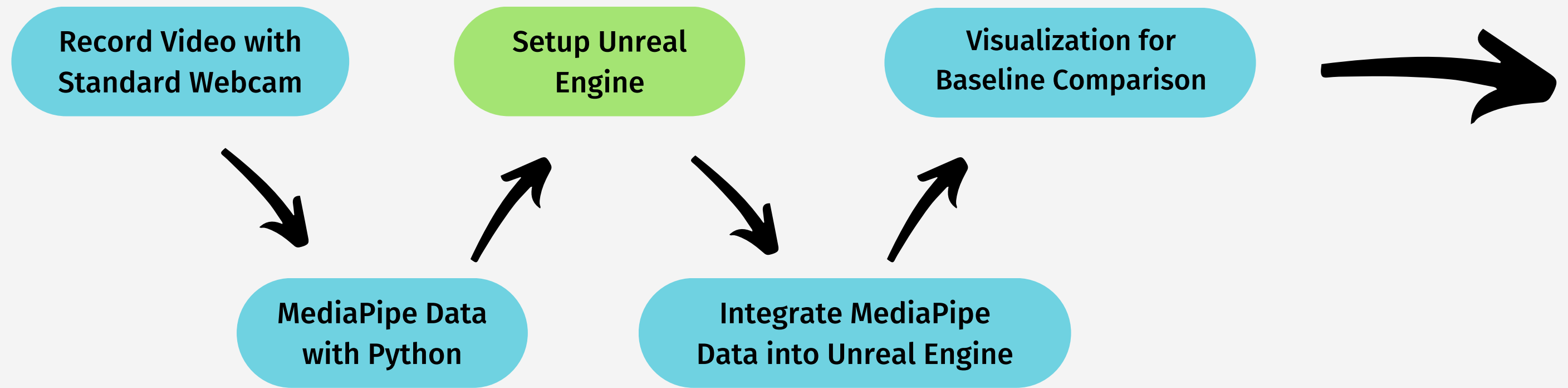
# Methodology.



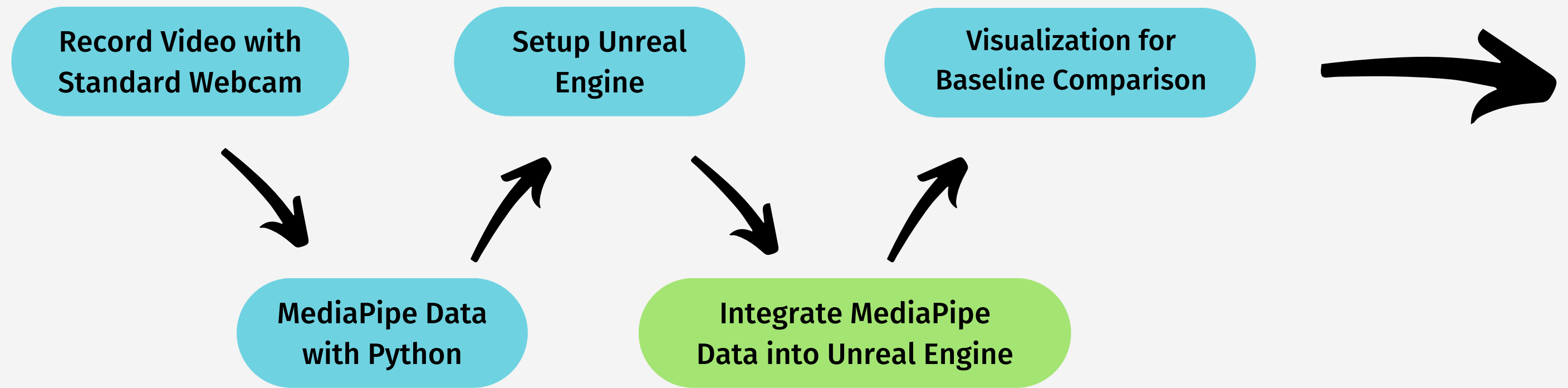
# Methodology.



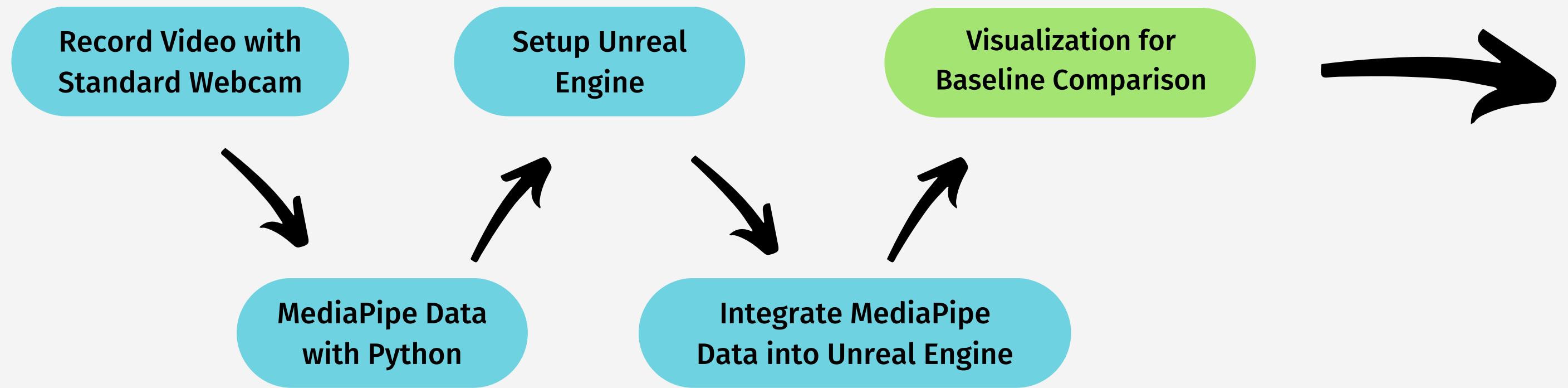
# Methodology.



# Methodology.



# Methodology.





# Methodology.



Apply Filtering  
Methods

Calculate Filtering  
Evaluation Metrics

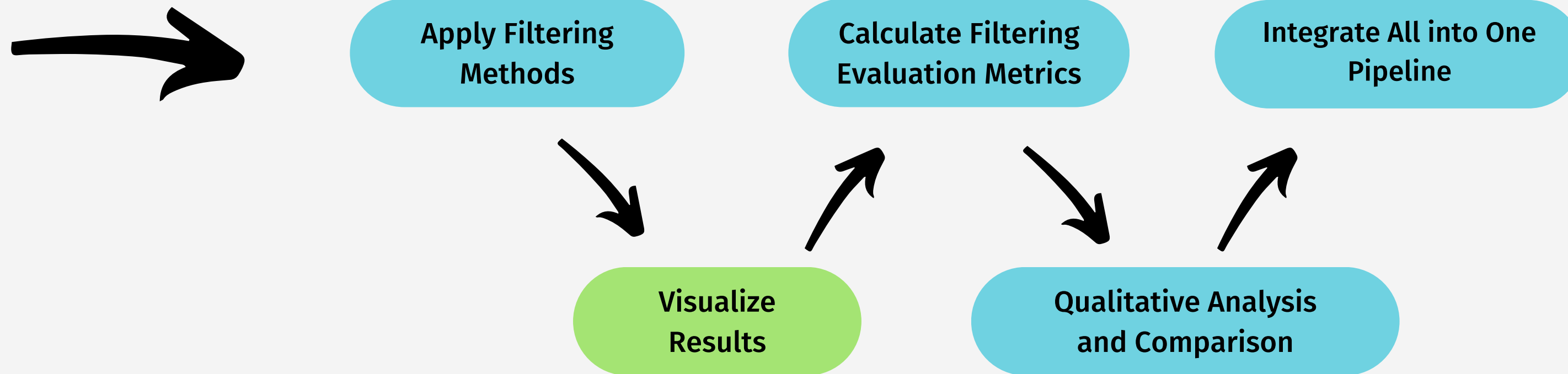
Integrate All into One  
Pipeline

Visualize  
Results

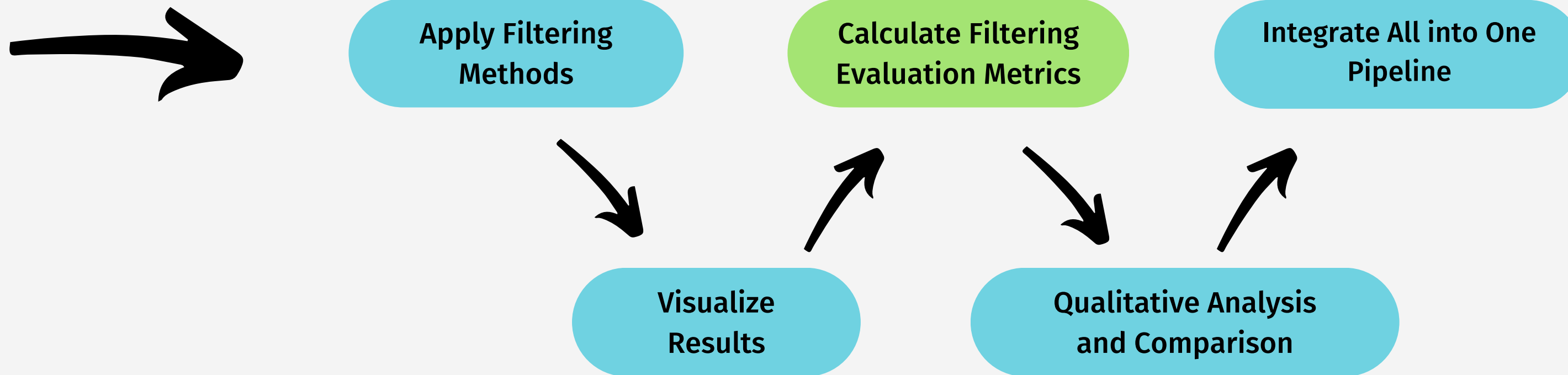
Qualitative Analysis  
and Comparison



# Methodology.

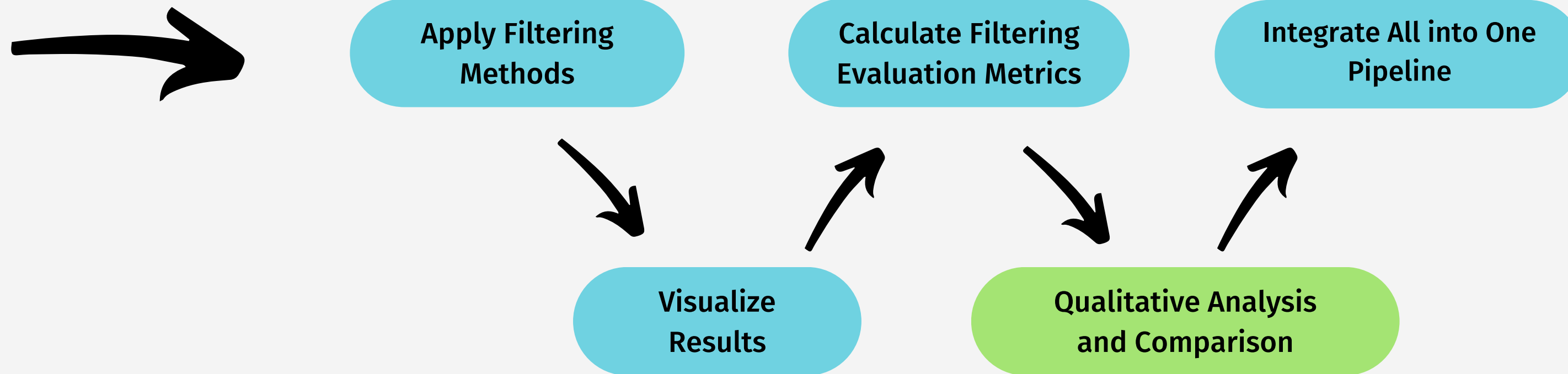


# Methodology.

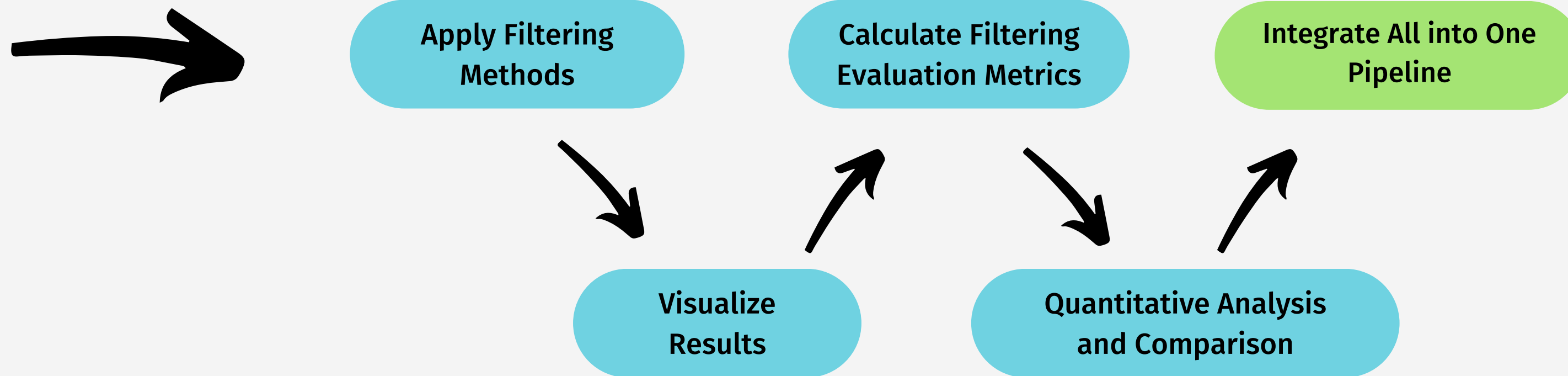




# Methodology.



# Methodology.



one branch for unfiltered, and one for filtered, and then merge for comparison, reduce text to fit all in one slides



**Thank You**

# Appendix

| ID  | TASK TITLE                                                                          | WEEK 1 |   |   |   |   | WEEK 2 |   |   |   |   | WEEK 3 |   |   |   |   | WEEK 4 |   |   |   |   | WEEK 5 |   |   |   |   | WEEK 6 |   |   |   |   | WEEK 7 |   |   |   |   | WEEK 8 |   |  |  |  | WEEK 9 |  |  |  |  | WEEK 10 |  |  |  |  | W |  |
|-----|-------------------------------------------------------------------------------------|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|---|---|---|--------|---|--|--|--|--------|--|--|--|--|---------|--|--|--|--|---|--|
|     |                                                                                     | M      | T | W | R | F | M      | T | W | R | F | M      | T | W | R | F | M      | T | W | R | F | M      | T | W | R | F | M      | T | W | R | F | M      | T | W | R | F | M      | T |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 1   | Literature Review (Hand Tracking, MediaPipe, Inverse Kinematics, Filtering Methods) |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 2   | MediaPipe Setup with Python for Data Generation                                     |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 3   | UE5 Environment Setup                                                               |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 3.1 | Control Rig Configuration on Humanoid Model                                         |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 3.2 | MediaPipe Data Integration into UE5                                                 |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 4   | Raw MediaPipe Data Visualization in UE5                                             |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 5   | Applying Filtering Methods (Standard Linear Kalman Filter and One Euro Filter)      |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 5.1 | Visualization of Filtering Results                                                  |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 5.2 | Filtering Metrics Evaluation                                                        |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 6   | Qualitative Analysis and Comparison                                                 |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 7   | Integrating All Components into One Pipeline                                        |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 8   | Testing                                                                             |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 9   | Finalize Project                                                                    |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 9.1 | Finalize Paper                                                                      |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |
| 9.2 | Prepare Paper Defense                                                               |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |   |   |   |        |   |  |  |  |        |  |  |  |  |         |  |  |  |  |   |  |

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